

Faria Ferdous

Dhaka 1000, Bangladesh

+880 1675 310643

faria.ferdous@du.ac.bd

Google Scholar

ORCID

Assistant Professor

Department of Physics, University of Dhaka

Research Profile

Experimental condensed matter physicist with research experience in magnetic and multiferroic materials, ferrite nanomaterials, functional oxide thin films, photocatalytic nanocomposites, and solar-cell transport layers. Research activities encompass materials synthesis, structural and optical characterization, magnetic and electrical property analysis, and investigation of structure-property relationships for energy and environmental applications. Published research includes peer-reviewed articles in materials science and applied physics journals, alongside an academic appointment as Assistant Professor in the Department of Physics at the University of Dhaka.

Research Interests

- Experimental condensed matter physics
- Magnetic, multiferroic, and functional oxide materials
- Ferrite nanoparticles, heterojunctions, and nanocomposites
- Metal-oxide thin films and charge-transport layers for perovskite solar cells
- Photocatalytic nanomaterials for environmental remediation
- Materials synthesis, thin-film deposition, and characterization

Education

- 2016 Master of Science, Department of Physics, University of Dhaka
CGPA: 3.92 out of 4.00; 2nd position
- 2015 Bachelor of Science, Department of Physics, University of Dhaka
CGPA: 3.65 out of 4.00; 6th position

Peer-Reviewed Publications

1. **Ferdous, F.**, Syed, I. M., & Hossain, A. K. M. “Promising magnetoelectric properties of $(x)\text{Ni}_{0.46}\text{Cu}_{0.14}\text{Zn}_{0.40}\text{Fe}_2\text{O}_4 + (1-x)\text{Ba}_{0.95}\text{Sr}_{0.05}\text{Zr}_{0.05}\text{Ti}_{0.95}\text{O}_3$ multiferroic composites.” *Journal of Materials Science: Materials in Electronics*, 32, 8514–8530, 2021. DOI: [10.1007/s10854-021-05480-x](https://doi.org/10.1007/s10854-021-05480-x).
2. Akter, S., Khan, M. N. I., **Ferdous, F.**, Das, H. N., & Syed, I. M. “Analysis of the influence of trivalent Cr^{3+} doping on the structural and electromagnetic properties of $\text{Cu}_{0.5}\text{Mg}_{0.5}\text{Cr}_x\text{Fe}_{2-x}\text{O}_4$ nanoferrites.” *AIP Advances*, 12(9), 095015, 2022. DOI: [10.1063/5.0110308](https://doi.org/10.1063/5.0110308).
3. Haider, M. A., Syed, I. M., Nahar, A., **Ferdous, F.**, & Khan, M. N. I. “Structural, electric and magnetic properties of $(1-x)\text{BaTiO}_3 + x\text{Cu}_{0.1}\text{Co}_{0.9}\text{Fe}_2\text{O}_4$ multiferroic composite.” *Bangladesh Journal of Physics*, 30(1), 43–56, 2023. DOI: [10.3329/bjphy.v30i1.79513](https://doi.org/10.3329/bjphy.v30i1.79513).

4. Akter, S., Khan, M. N. I., **Ferdous, F.**, Das, H. N., Alam, M. K., Rahman, M. A., Hasan, T., & Syed, I. M. “Unveiling the role of sintering temperatures in the physical properties of Cu-Mg ferrite nanoparticles for photocatalytic application.” *Heliyon*, 10, e40771, 2024. DOI: [10.1016/j.heliyon.2024.e40771](https://doi.org/10.1016/j.heliyon.2024.e40771).
5. Jahan, S., Sayem, M. A., Uddin, M. K., **Ferdous, F.**, Jahan, R. A., & Syed, I. M. “Remarkable photocatalytic performance and recyclable pathway of cleansing wastewater using ZnO-rGO nanocomposite.” *AIP Advances*, 15, 075343, 2025. DOI: [10.1063/5.0275565](https://doi.org/10.1063/5.0275565).

Manuscript in Preparation

1. Enhancing Ion Accessibility in MoO₃/rGO Networks through Interfacial Defect Modulation for High-Performance Symmetric Aqueous Supercapacitors.

Research Experience and Ongoing Projects

- **Principal Investigator, University Grants Commission (UGC), Bangladesh | 2025–2026**
Study of Graphene Oxide-Based Semiconducting Nanocomposites for Ultra-Fast Photocatalytic Degradation in Wastewater Management
Project ID: 21463331
- Fabrication of Metal Oxide Thin Films by Chemical Bath Deposition for Efficient Electron Transport Layers in Perovskite Solar Cells
- Structural, Magnetic, Dielectric, Optical, and Photocatalytic Study of Cu-Substituted Manganese Ferrite/CeO₂ Heterojunction Nanocomposites
- Development of Calcium Nitrate-Doped NiO Hole Transport Layers by Spin Coating for Solar Cell Applications

Research Skills and Techniques

- Thin-film deposition: chemical bath deposition (CBD) and spin coating
- Ferrite nanoparticle and oxide/nanocomposite synthesis
- Structural characterization and phase analysis using X-ray diffraction (XRD)
- Magnetic characterization using a Vibrating Sample Magnetometer (VSM)
- Optical and spectroscopic characterization using UV-Vis and FTIR spectroscopy
- Dielectric and impedance measurements
- Photocatalytic performance evaluation and wastewater degradation studies
- Perovskite solar cell fabrication and transport-layer development

Computational and Software Skills

- Programming and Scientific Computing: C/C++, Maple, MATLAB
- Software Tools: OriginLab, FullProf, ImageJ/Fiji, Microsoft Office, LaTeX.

Academic Appointments

2023–Present	Assistant Professor, Department of Physics, University of Dhaka Dhaka, Bangladesh
2019–2023	Lecturer, Department of Physics, University of Dhaka Dhaka, Bangladesh
2018–2019	Lecturer in Physics, Primeasia University Dhaka, Bangladesh

Awards and Fellowships

2017–2018	National Science and Technology Fellowship, Ministry of Science and Technology, Bangladesh.
2015	Merit Scholarship, Directorate of Secondary and Higher Education, Bangladesh.

Mentoring

2021–2024	Student Advisor, Department of Physics, University of Dhaka.
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Teaching Experience

- Teaching undergraduate physics courses including Optics, Nuclear Physics, Electricity and Magnetism, and Mechanics.
- Supervising laboratory sessions, student projects, and experimental work.
- Supporting departmental academic activities and undergraduate mentoring.

Academic Service and Outreach

- Member, Organizing Committee, International Conference on Physics 2026.
- Member, Organizing Committee, Sultan Ahmed Memorial Conference: Recent Advances in Physics, 2024.
- Examiner, 7th Bangladesh Physics Olympiad.

Professional Memberships

- Life Member, Bangladesh Physical Society (LM R0019).

Conference Presentations

- Oral Presentation, International Conference on Physics 2018.